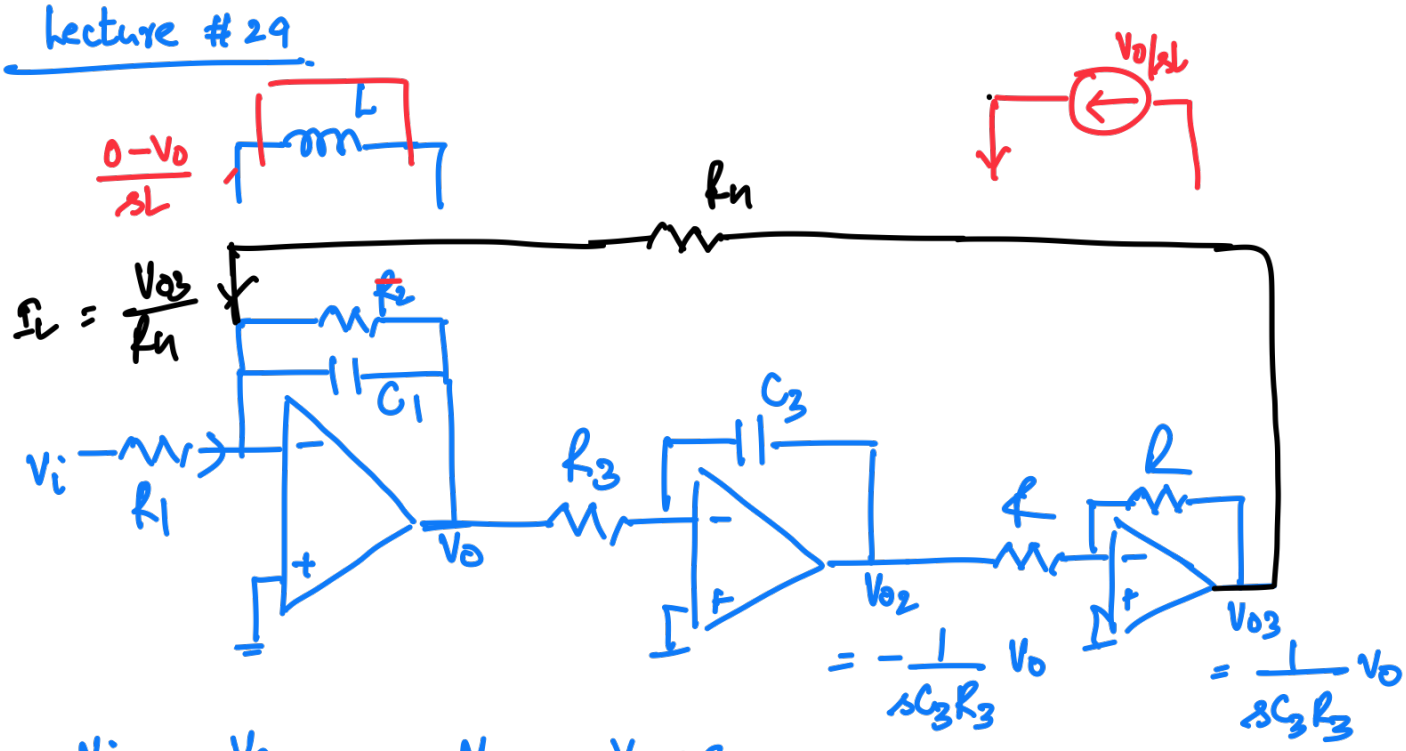
Lecture # 29



$$\frac{V_i}{R_1} + \frac{V_0}{sL} = -\frac{V_0}{R_2} - V_0 sC_1$$

$$I_L = \frac{V_0}{sC_3R_3R_4} \Rightarrow L = C_3R_3R_4$$

$$\frac{V_0}{V_i} = \frac{-\frac{sL}{R_2}\cdot\frac{R_2}{R_1}}{s^2LC_1 + \frac{sL}{R_2} + 1}$$

$$\frac{V_i}{R_1} + \frac{V_0}{sC_3R_3R_4} = -\frac{V_0}{R_2} - V_0 sC_1$$

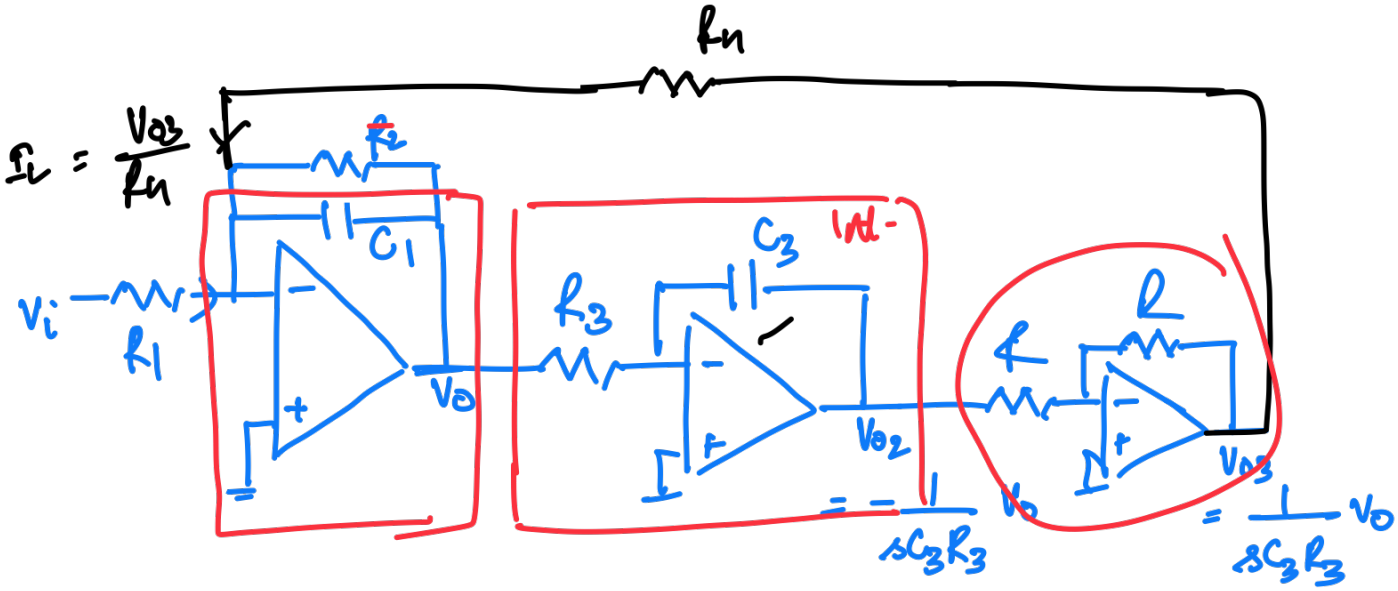
$$V_0\left(\frac{1}{R_2} + sC_1 + \frac{1}{sL}\right) = -\frac{V_i}{R_1}$$

$$\frac{V_0}{V_i} = \frac{-\frac{R_2}{R_1}\cdot\frac{sL}{R_2}}{s^2C_1L + \frac{sL}{R_2} + 1}$$

$L = C_3R_3R_4$

$$\frac{V_0}{V_i} = -\frac{R_2}{R_1}\,\frac{s\cdot\frac{C_3R_3R_4}{R_2}}{s^2C_1C_3R_3R_4 + s\frac{C_3R_3R_4}{R_2} + 1}$$

$$\frac{V_{02}}{V_i} = \frac{+\frac{R_2}{R_1}\times\frac{R_4}{R_2}}{(\quad)}$$



- Opamp, R, C (Active-RC)

$$\frac{V_o}{V_i} = \frac{s/\omega_p Q_p}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1} \cdot A_0$$

$$V_o \left[\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1 \right] = A_0 \cdot \frac{s}{\omega_p Q_p} V_i$$

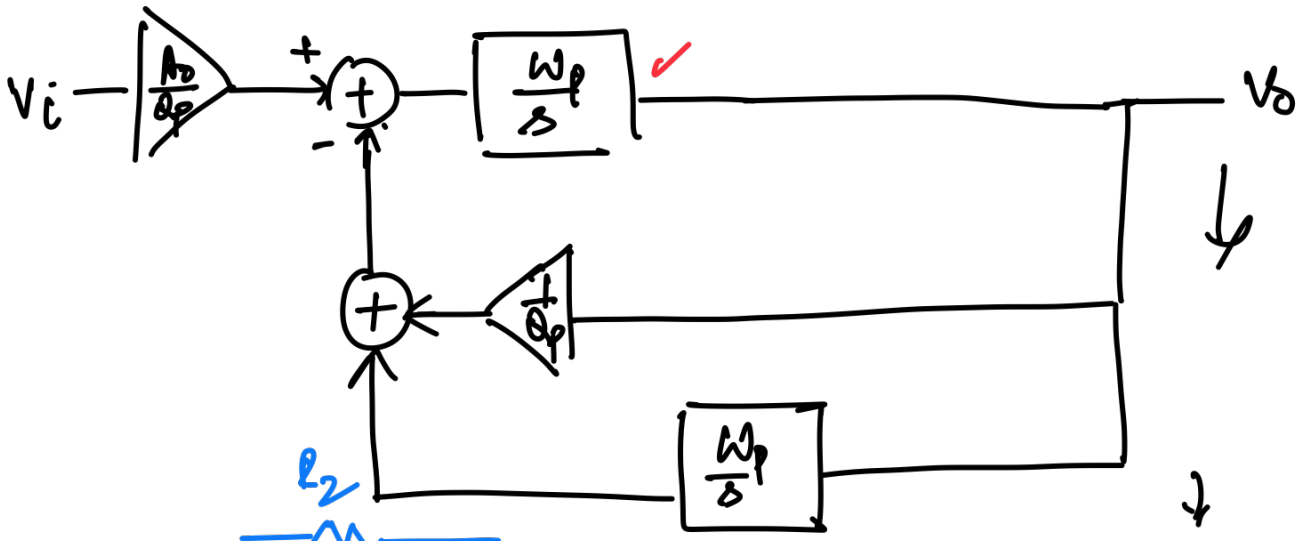
$$V_o \left[\frac{1}{\omega_p^2} + \frac{1}{s} \cdot \frac{1}{\omega_p Q_p} + \frac{1}{s^2} \right] = \frac{A_0}{\omega_p Q_p} \frac{1}{s} \cdot V_i$$

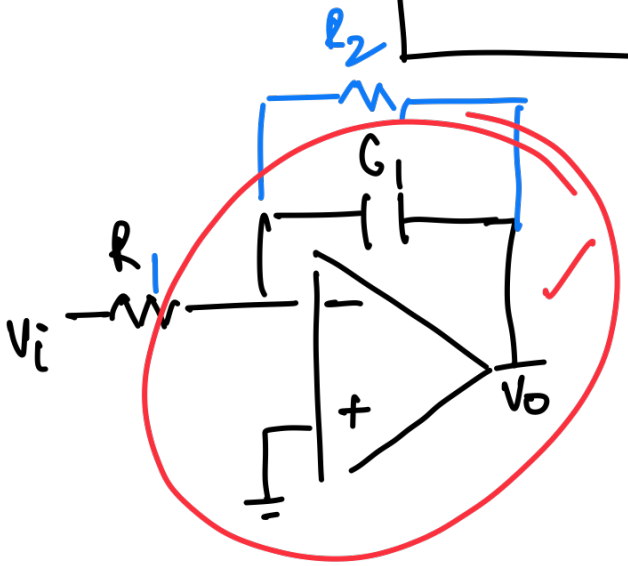
$$\frac{V_o}{s} \frac{1}{\omega_p Q_p} - \frac{A_0}{\omega_p Q_p} \frac{1}{s} V_i + \frac{V_o}{s^2} = \frac{-V_o}{\omega_p^2}$$

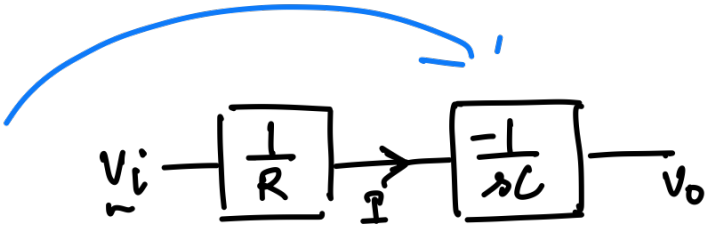
$$V_o = -\frac{V_o}{s} \frac{\omega_p}{Q_p} + \frac{A_0}{Q_p} \frac{\omega_p}{s} V_i - \frac{V_o}{s^2} \omega_p^2$$

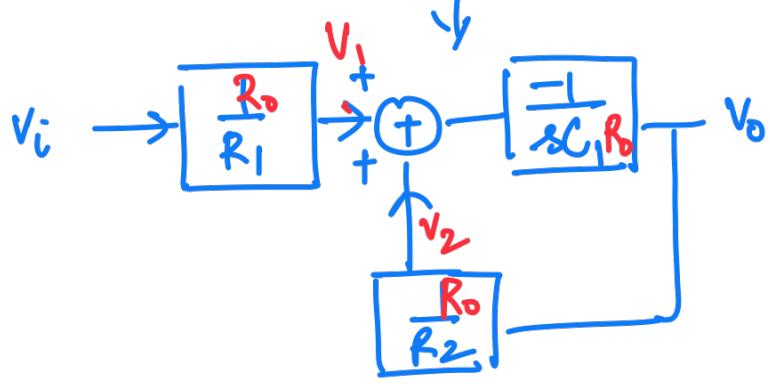
$$= \left(\frac{A_0}{Q_p} V_i - \frac{V_o}{Q_p} \right) \frac{\omega_p}{s} - V_o \cdot \frac{\omega_p^2}{s^2}$$

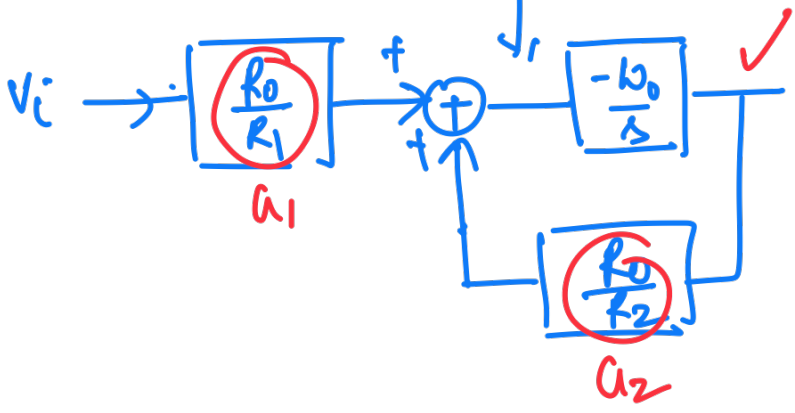
$$V_o = \left[\left(\frac{A_0}{Q_p} V_i - \frac{V_o}{Q_p} \right) - V_o \cdot \frac{\omega_p}{s} \right] \frac{\omega_p}{s}$$

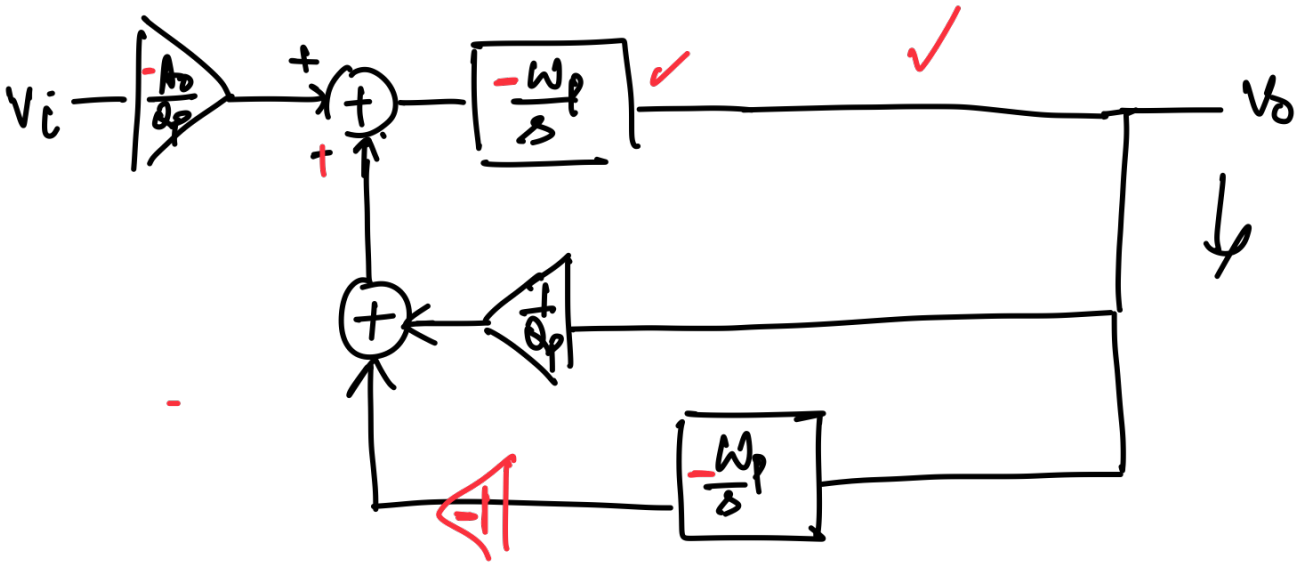


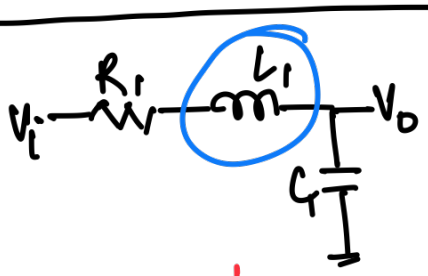




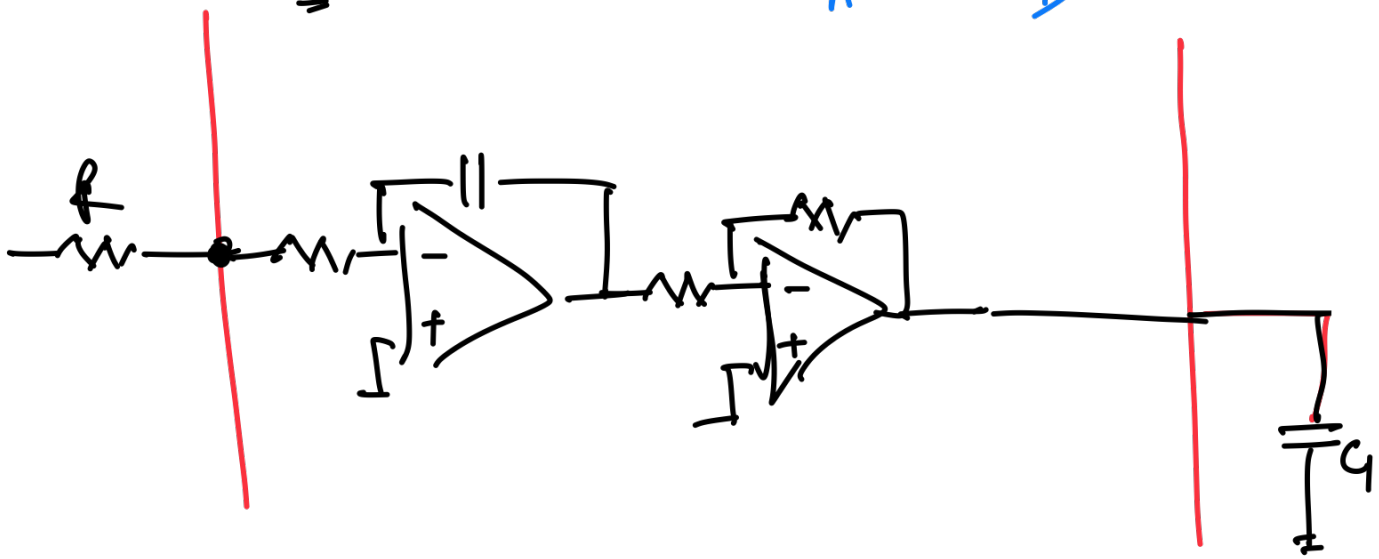












$$V_o = \frac{1/sC_1 \quad V_i}{\frac{1}{sC_1} + R_1 + sL_1} = \frac{1}{s^2 L_1 C_1 + s C_1 R_1 + 1} V_i$$

$$= \frac{V_i}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

$$V_o\left(\frac{s^2}{\omega_p^2}+\frac{s}{\omega_p Q_p}+1\right) = V_i$$

$$V_o\left(1+\frac{\omega_p}{s}\cdot\frac{1}{Q_p}+\frac{\omega_p^2}{s^2}\right) = V_i\,\frac{\omega_p^2}{s^2}$$

$$V_o = \left((V_i - V_o)\frac{\omega_p}{s} - \frac{V_o}{Q_p}\right)\frac{\omega_p}{s}$$

